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Activity Log Analyzer in Python

1. Introduction

This assignment focuses on developing an Activity Log Analyzer using Python. The goal of the analyzer is to process activity logs of students and analyze their online activities such as applications used, websites visited, and the time spent on each activity. Python's built-in data structures like lists, dictionaries, and sets are used to store and process the data efficiently.

2. Activity Log Structure

Each activity record contains the following fields:

```
{ "user": str, "action": str, "duration": float }
```

user: Roll number of the student
action: Online activity performed (apps, websites, etc.)
duration: Time spent on that activity

3. Python Implementation

```
from typing import List, Dict, Set
from collections import defaultdict
from functools import reduce

def total_time_per_user(logs: List[Dict]) -> Dict[str, float]:
    user_times = defaultdict(float)
    for log in logs:
        user_times[log["user"]] += log["duration"]
    return dict(user_times)

def most_active_users(logs: List[Dict], k: int) -> List[str]:
    totals = total_time_per_user(logs)
```

```
    return [user for user, _ in sorted(totals.items(), key=lambda x: x[1], reverse=True)[:k]]

def unique_actions(logs: List[Dict]) -> Set[str]:
    return {log["action"] for log in logs}

def total_activity_time(logs: List[Dict]) -> float:
    return reduce(lambda acc, log: acc + log["duration"], logs, 0.0)
```

4. Complexity Analysis

Time Complexity for computing top K users:

If there are n activity records, calculating total time per user takes $O(n)$. Sorting users takes $O(u \log u)$, where u is the number of unique users. So the total time complexity is $O(n + u \log u)$.

Space Complexity:

The dictionary storing total time per user requires $O(u)$ space where u is the number of unique users. The set storing unique actions requires $O(a)$ space where a is the number of unique actions. Therefore the overall space complexity is $O(u + a)$.

5. Conclusion

The Activity Log Analyzer demonstrates how Python built-in containers can be used to efficiently analyze user activity data. By using dictionaries, sets, list comprehensions, and functions such as `sorted()` and `reduce()`, the program performs analysis efficiently and in a clean Pythonic style.